

Annex — Formula & Implementation Compendium

Weather-Aware SIKD over Tropical Satellite Free-Space Optical Channels

Every formula, coefficient, and quantity behind the results and the interactive study site.

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Abstract

This annex consolidates the complete mathematical machinery of the thesis: the SIKD signal and channel model, the dual-threshold receiver, the three-state weather model, the coverage geometry, the two-level scheduling formulation, the inter-satellite-link relay, and the key/data power-split, together with every coefficient and a glossary of all physical quantities (value, typical range, rationale). Sections A–H are the curated derivations (source: 05_Code_v3/modules and the self-study guides in Tu-hoc/SIKD_Satellite_QKD); Sections I–J are generated directly from the data served by the interactive site (<https://tatcataittn.github.io/for-ttn-USTH/Thesis-M1>), so every online implementation is indexed here.

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1 Master parameter and coefficient table

All defaults are the corrected v3 values ($m_K = 0.15$).

Symbol	Value	Meaning
λ	1550 nm	carrier wavelength (telecom C-band)
θ_C	10 μ rad	beam divergence half-angle
a_R	0.05 m	receiver aperture radius
H_S	550 km	satellite altitude (Starlink Shell-1)
H_{atm}	20 km	effective atmospheric height
R_E	6371 km	mean Earth radius
W	21 m/s	high-altitude wind (Hufnagel–Valley)
$C_n^2(0)$	$1.7 \times 10^{-14} \text{ m}^{-2/3}$	ground turbulence strength
R_e	0.9 A/W	InGaAs PIN responsivity
T	280 K	receiver temperature
R_L	1 k Ω	load resistance
R_b	1 Gbps	raw bit rate ($\Delta f = R_b/2$)
P_{bg}	0	background power (night operation)
Iso	15 dB	subcarrier filter isolation
P_T	1 W (30 dBm)	transmit optical power
m_K	0.15	key-channel modulation index
m_D	0.5	data-channel modulation index
ζ_{scale}	2.0	dual-threshold scale factor
N_{GH}	20	Gauss–Hermite quadrature order
α, ρ	0.509, 0.63	tropical rain law (Koné 2024, Abidjan)
Cloud threshold	85%	cloud-cover outage threshold
Rain-hour threshold	0.1 mm/h	measurable-precipitation threshold
Record	2015–2024 (10 yr)	ERA5/ERA5-Land, 8 cities
Handover cost	30 s	real acquisition time per re-assignment
ISL hop latency	30 s	conservative on-board relay overhead
QBER _{max}	11%	BB84 Shor–Preskill security threshold

2 SIKD signal and FSO channel model

Received SCM photocurrent (direct detection):

$$i_r(t) = \frac{1}{2} R_e P_T h(t) [1 + m_K s_K(t) + m_D s_D(t)] + n(t)$$

One 1550 nm carrier multiplexes the quantum key (f_1, m_K) and classical data (f_2, m_D); both share one photodetector, so s_D returns as cross-talk.

Composite channel: $h = h_g \cdot h_l \cdot h_a$ (geometric $\times$ atmospheric $\times$ turbulence).

2.1 Geometric spreading

$$w_0 = \frac{\lambda}{2\theta_C}, \quad w_L = w_0 \sqrt{1 + \left(\frac{L\lambda}{\pi w_0^2} \right)^2}, \quad h_g = [\text{erf}(\nu_R)]^2, \quad \nu_R = \sqrt{\frac{\pi}{2}} \frac{a_R}{w_L}$$

Spherical slant range (elevation $\varepsilon = 90^\circ - \zeta$):

$$L = \sqrt{R_E^2 \sin^2 \varepsilon + 2R_E H_S + H_S^2} - R_E \sin \varepsilon$$

v3 fixes: ν_R coefficient $\sqrt{2} \rightarrow \sqrt{\pi/2}$; slant range flat-Earth $\rightarrow$ spherical (992.8 km, not 1100 km, at $\varepsilon = 30^\circ$). h_g exceeds 30 dB alone.

2.2 Atmospheric attenuation (Beer–Lambert)

$$h_l = \exp[-(\sigma_{\text{clear}} + \sigma_{\text{rain}})L_{\text{atm}}], \quad L_{\text{atm}} = \frac{H_{\text{atm}} - H_U}{\cos \zeta}$$

Clear air (Kruse/Kim): $\sigma_{\text{clear}} = \frac{3.912}{V} \left(\frac{\lambda}{550}\right)^{-q(V)}$, with $q(V) = 1.6$ ($V > 50$); 1.3 ($6 < V \leq 50$); $0.16V + 0.34$ ($1 < V \leq 6$); $V - 0.5$ ($0.5 < V \leq 1$); 0 else.

Rain (tropical): $\sigma_{\text{rain}} = \alpha R^\rho / 4.343$, $\alpha = 0.509$, $\rho = 0.63$; $4.343 = 10 / \ln 10$ (dB $\rightarrow$ Neper). Independent extinction mechanisms add in the exponent; $V = 15 - 10P_{\text{cloud}}$ is derived (no ERA5 visibility variable).

2.3 Turbulence (weak, log-normal)

Hufnagel–Valley 5/7 profile:

$$C_n^2(h) = 0.00594 \left(\frac{W}{27}\right)^2 (10^{-5}h)^{10} e^{-h/1000} + 2.7 \times 10^{-16} e^{-h/1500} + C_n^2(0) e^{-h/100}$$

Rytov variance and log-amplitude variance:

$$\sigma_R^2 = 2.25 k^{7/6} \sec^{11/6}(\zeta) \int_{H_U}^{H_S} C_n^2(h) (h - H_U)^{5/6} dh, \quad \sigma_X^2 = \sigma_R^2 / 4$$

Fading: $h_a = \exp(2X)$, $X \sim \mathcal{N}(-\sigma_X^2, \sigma_X^2)$ (so $E[h_a] = 1$); log-normal while $\sigma_R^2 < 0.3$, else Gamma–Gamma. v3 fix: Rytov coefficient $0.56 \rightarrow 2.25$ (plane-wave slant path). The slant path stays weak, so log-normal holds throughout.

3 Receiver: noise, sifting, QBER and secret key rate

Noise budget ($\Delta f = R_b/2$):

$$\sigma_N^2 = \underbrace{2qR_e \left(\frac{1}{2}P_T h_g h_l\right) \Delta f}_{\text{shot}} + \underbrace{\frac{4k_B T}{R_L} \Delta f}_{\text{thermal}} + \underbrace{2qR_e P_{bg} \Delta f}_{\text{background}} + \underbrace{\frac{1}{8} (R_e P_T m_D h_g h_l)^2 10^{-\text{Iso}/10}}_{\text{cross-talk}}$$

v3 fixes: shot noise uses the total DC photocurrent (no m_K); cross-talk denominator $/2 \rightarrow /8$. Cross-talk $\propto \text{power}^2$, so it dominates a strong link ($0.5\text{--}290\times$ thermal across elevation).

Dual-threshold decision ($i_{\text{mean}} = \frac{1}{2}R_e G_k P_T m_K h_g h_l$):

$$d_0 = -i_{\text{mean}} - \zeta_{\text{scale}} \sigma_N, \quad d_1 = +i_{\text{mean}} + \zeta_{\text{scale}} \sigma_N$$

Sift probability and QBER (20-point Gauss–Hermite over fading):

$$P_{\text{sift}} = \frac{1}{2}(P_{00} + P_{11} + P_{10} + P_{01}), \quad \text{QBER} = \frac{\frac{1}{2}(P_{10} + P_{01})}{P_{\text{sift}}}$$

Secret key rate (Shor–Preskill / Devetak–Winter, $f_{EC} = 1$):

$$\text{SKR}_{\text{norm}} = P_{\text{sift}} [1 - (1 + f_{EC})H_2(\text{QBER}) - \chi_E] = P_{\text{sift}} [1 - 2H_2(\text{QBER}) - \chi_E], \quad \text{SKR} = \text{SKR}_{\text{norm}} R_b$$

with $H_2(p) = -p \log_2 p - (1-p) \log_2 (1-p)$, $= 0$ at QBER $\approx 11\%$. v3 fix: the error-correction leakage term $f_{EC}H_2(\text{QBER})$ was missing; it makes SKR vanish at the BB84 threshold. Baseline $\chi_E = 0$.

Classical data BER: $\text{BER}_{CC} = \frac{1}{\sqrt{\pi}} \sum_k w_k Q\left(\frac{A_0 h_{a,k}}{\sigma_{N,D}}\right)$, $A_0 = G_d R_e P_T m_D h_g h_l$.

4 Three-state weather model and availability

$$p_{\text{rain}} = \min(f_{\text{rain}}, 1 - P_{\text{cloud}}), \quad p_{\text{clear}} = \max(0, 1 - P_{\text{cloud}} - p_{\text{rain}}), \quad p_{\text{cloud}} = P_{\text{cloud}}$$

$$\text{SKR}_{\text{eff}} = p_{\text{clear}}\text{SKR}_c + p_{\text{rain}}\text{SKR}_r, \quad A = p_{\text{clear}} + p_{\text{rain}} = 1 - P_{\text{cloud}}$$

Rain rate: $R_{\text{mm/h}} = R_{\text{month}} / (30 \cdot 24 \cdot f_{\text{rain}})$.

Statistics: P_{cloud} = fraction of hours with cloud cover $\geq 85\%$; f_{rain} = fraction with precip ≥ 0.1 mm/h; inter-city Pearson correlation of daily cloud; joint-clear probability from the joint record. [Cloud is a binary outage](#); A is invariant to f_{rain} . Availability spans 10.7% (Kuala Lumpur, Nov) to 96.5% (Bangkok, Feb). Best combined FSO window: mid-morning ($\sim 05:00$ – $10:00$).

5 Coverage geometry and city-pair feasibility

$$R_{\text{cov}} = R_E \left[\arccos\left(\frac{R_E}{R_E + H_S} \cos \varepsilon\right) - \varepsilon \right], \quad \text{DUAL if } d \leq 2R_{\text{cov}} \text{ else store-and-forward}$$

Great-circle distance (haversine): $d = 2R_E \arcsin \sqrt{\sin^2 \frac{\Delta\varphi}{2} + \cos \varphi_1 \cos \varphi_2 \sin^2 \frac{\Delta\lambda}{2}}$.

Off-nadir angle: $\sin \eta = R_E \cos \varepsilon / (R_E + H_S)$. Slant range: $\sqrt{(R_E + H_S)^2 - (R_E \cos \varepsilon)^2} - R_E \sin \varepsilon$.

Directed store-and-forward latency: median time from a pass over i to the same satellite's next pass over j (asymmetric). $R_{\text{cov}} = 793$ km (30°), 573 km (40°). The 28 ASEAN pairs split 14 DUAL / 14 store-and-forward — a geographic ceiling.

6 Weather-aware scheduling and pair-key allocation

Level 1 — matching (Hungarian, per step t):

$$\max_x \sum_{g,s} w_{g,s}(t) x_{g,s}(t) - c_h \sum_g \mathbf{1}[\text{handover}], \quad \sum_s x_{g,s} \leq 1, \quad \sum_g x_{g,s} \leq 1, \quad x = 0 \text{ if not visible}$$

weight $w_{g,s} = \text{SKR}_{\text{eff}}$.

Level 2 — pair key (trusted relay, satellite broadcasts $k_A \oplus k_B$):

$$K_q = \min \left(\sum_{p: \text{station}(p)=A} k_p y_{p,q}, \sum_{p: \text{station}(p)=B} k_p y_{p,q} \right), \quad k_p = w \cdot (t_{\text{set}} - t_{\text{rise}})$$

Strategies: ALG-0 (sticky elevation-priority baseline; ALG-0-feasible resolves double-booking); ALG-1 (weather-aware Hungarian matching, blind vs aware); ALG-2 (pass-to-pair allocation: greedy deficit rule, MILP/HiGHS, or max-min water-filling). [Fair comparison at equal resources over 620 real days](#): matching gain $+2.47 / +2.48\%$; weather-info gain $-0.29 / -0.23\%$; pair-starvation 91.6–96.5%. Handover cost carried across segments (median 3.3 s).

7 Inter-satellite-link relay and key/data power-split

ISL multi-hop relay. BFS hop distance on the static “+Grid” mesh; a drop-off pass is feasible if $t_{\text{rise}} \geq t_{\text{pick}} + n_{\text{hops}} \cdot 30$ s. *time-optimal* = earliest feasible; *capacity-optimal* = highest-elevation (highest-SKR) within $3\times$ the time-optimal latency. Replaces single-hop store-and-forward (worst case ~ 711 min) with 2.9–4.1 min.

Power-split. Two-way cross-talk $\sigma_{\text{CT}}^2 = (R_e P_T m_{\text{leak}} h_g h_l)^2 10^{-\text{Iso}/10} / 8$ ($m_{\text{leak}} = m_D$ into key, m_K into data). Data goodput $= R_b(1 - \text{BER})$; constraint $m_K + m_D \leq 1$. Pareto frontier over (m_K, m_D) ; adaptive split picks the smallest m_K with $\text{SKR} \cdot 10^3 \cdot T_{\text{pass}} \geq K_{\text{req}}$. [Raising \$m_D\$ improves data throughput but \$\sigma_{\text{CT}}^2 \propto m_D^2\$ raises key QBER — the central SIKD coupling.](#)

8 Glossary of physical quantities

For each quantity: meaning, this project's value, the typical range and why it is chosen, and how that range differs across fields (as served by the interactive site).

- **λ — Optical wavelength.** The carrier light's wavelength; sets diffraction, atmospheric scattering, and detector response.
This project: 1550 nm (telecom C-band) throughout this project. **Typical range:** Free-space optics: ~800 nm, ~1064 nm, or 1550 nm. This project fixes 1550 nm.
Why: 1550 nm is eye-safe at higher power, sits in the atmosphere's low-loss window, scatters less off aerosols than visible light (Kruse/Kim exponent), and reuses mature, cheap telecom lasers/detectors/fibre.
Across fields: Telecom/satellite FSO almost always pick 1550 nm for those reasons. General optics/astronomy span UV→IR freely; lidar often uses 532/1064 nm. Quantum labs sometimes prefer shorter wavelengths where single-photon detectors are most efficient — a tension this direct-detection design sidesteps.
- **θ_C — Beam divergence half-angle.** How fast the transmitted laser beam spreads; smaller means a tighter, farther-reaching beam.
This project: 10 μ rad. **Typical range:** Satellite FSO: ~1–50 μ rad.
Why: 10 μ rad keeps geometric spreading loss manageable at 550 km, but a tighter beam demands finer pointing (PAT), so it trades against pointing budget.
Across fields: Satellite laser-comm lives in the microradian regime because links are thousands of km. RF antennas think in degrees (milliradians+); a laser pointer is ~1 mrad — 100× wider than this.
- **w_0 — Beam waist.** The beam's narrowest radius, at the transmitter: $w_0 = \lambda/(2\theta_C)$.
This project: ≈ 0.0775 m (7.75 cm). **Typical range:** Centimetre-scale for these parameters.
Why: Fixed by λ and θ_C ; a smaller divergence gives a larger waist and slower far-field spreading.
Across fields: A derived quantity specific to Gaussian-beam optics; RF engineering has no direct analogue.
- **w_L — Beam radius at the receiver.** The spot radius after propagating the slant range L; grows ~linearly with distance.
This project: ≈ 3.5 m (zenith) to ≈ 6.3 m (30° elevation). **Typical range:** Metres, at LEO ranges.
Why: Because w_L (metres) $\gg$ aperture a_R (5 cm), only $\sim(a_R/w_L)^2$ of the power is collected — the origin of the >30 dB geometric loss.
Across fields: This metre-scale spot over a centimetre aperture is why FSO link budgets are dominated by geometry, unlike fibre (no spreading) or RF (much wider beams but larger antennas).
- **a_R — Receiver aperture radius.** Radius of the ground telescope collecting the beam.
This project: 5 cm (10 cm diameter). **Typical range:** Satellite FSO ground optics: a few cm (CubeSat-class, e.g. TBIRD ~0.2 m) up to ~1 m (Micius ground station).
Why: A small aperture is cheap/light and lets you deploy many stations, but collects a smaller fraction of the already-weak key signal — an unresolved SWaP trade-off here.
Across fields: Astronomy uses metre-to-10-metre apertures for faint light; this project deliberately picks the small, cheap end to argue a multi-station network.
- **h_g — Geometric spreading loss.** Fraction of transmitted power reaching the aperture, from beam spreading: $h_g = [\text{erf}(v_R)]^2$.
This project: ≈ -34 dB (zenith) to ≈ -39 dB (30° elevation). **Typical range:** Tens of dB (very lossy) for LEO FSO.
Why: It scales $\sim 1/L^2$, so lower passes (longer range) are much worse; it is the single largest term in the link budget.
Across fields: Uniquely severe in free-space optics; fibre has none, and RF free-space path loss, while large, is offset by high antenna gains.
- **h_l — Atmospheric transmission.** Beer-Lambert transmission through the air: $h_l = \exp(-\sigma_{\text{total}} \cdot L_{\text{atm}})$.
This project: ≈ -9 dB (clear, zenith) to ≈ -18 dB (clear, 30°); ~ -100 dB in heavy rain. **Typical range:** A few dB (clear) to link-killing (rain/fog).
Why: Set by visibility and rain rate along the slant path; low elevation lengthens the path ($1/\cos \zeta$).

Across fields: Remote sensing / atmospheric science model exactly this extinction. In this project cloud is treated as a separate binary outage, not folded into h_1 .

- **σ_{clear} — Clear-air extinction coefficient.** Per-km attenuation from aerosols/molecules (Kruse/Kim): $\sigma = (3.912/V)(\lambda/550)^{(-q)}$.
This project: $\approx 0.10 \text{ km}^{-1}$ at $V = 10 \text{ km}$, 1550 nm . **Typical range:** $\sim 0.05 \text{ km}^{-1}$ (very clear) to several km^{-1} (fog).
Why: Inversely tied to meteorological visibility V ; the wavelength factor makes 1550 nm scatter less than visible.
Across fields: A staple of atmospheric optics / lidar; identical machinery to remote-sensing atmospheric correction.
- **σ_{rain} — Rain extinction coefficient.** Per-km attenuation from raindrops: $\beta = \alpha R^\rho$ (dB/km), $\sigma_{\text{rain}} = \beta/4.343$.
This project: $\alpha = 0.509$, $\rho = 0.63$ (tropical, Koné 2024, Abidjan). **Typical range:** 0 (no rain) to $\sim 1.4 \text{ km}^{-1}$ at 50 mm/h .
Why: The sub-linear exponent ($\rho < 1$) reflects Mie scattering by large drops; calibrated on real tropical measurements, not textbook constants.
Across fields: Tropical FSO needs region-specific coefficients; temperate ITU rain models differ. RF rain fade (Ka-band) uses a related but distinct power law.
- **R — Rain rate.** Instantaneous rainfall intensity during raining hours.
This project: Converted from monthly totals: $R = R_{\text{month}}/(30 \cdot f_{\text{rain}})$. **Typical range:** Light drizzle $\sim 0.1 \text{ mm/h}$ to tropical downpour $30\text{--}50+$ mm/h.
Why: Tropical rain is intermittent, so monthly totals must be concentrated into the hours it actually rains (f_{rain}).
Across fields: Meteorology/hydrology quantity; FSO and Ka-band RF both care because both suffer rain fade, unlike lower RF bands.
- **V — Meteorological visibility.** Horizontal distance at which contrast drops to a threshold; a proxy for aerosol load.
This project: Derived here as $V = 15 - 10 \cdot P_{\text{cloud}}$ (no historical visibility record exists). **Typical range:** $\geq 10\text{--}50 \text{ km}$ (clear) down to $< 1 \text{ km}$ (fog).
Why: It directly sets σ_{clear} ; a derived estimate is used because ERA5 has no visibility variable.
Across fields: A standard weather-station output; here it is inferred, a stated modelling limitation.
- **H_S — Satellite altitude.** Height of the LEO satellite above the ground.
This project: 550 km (Starlink Shell-1). **Typical range:** LEO $\sim 300\text{--}2000 \text{ km}$; this project fixes 550 km .
Why: The channel model is parameterised for one altitude; mixing shells would need altitude-dependent tables.
Across fields: 550 km is the modern mega-constellation sweet spot (low latency, manageable drag). GEO comms sit at $35\,786 \text{ km}$; the ISS at $\sim 400 \text{ km}$.
- **ζ / ϵ — Zenith / elevation angle.** ζ is measured from straight up; elevation $\epsilon = 90^\circ - \zeta$ is measured from the horizon.
This project: Operating range $\epsilon = 30^\circ\text{--}90^\circ$ ($\zeta = 0^\circ\text{--}60^\circ$). **Typical range:** Elevation masks of $20^\circ\text{--}40^\circ$ are common.
Why: Low elevation lengthens both the geometric range and the atmospheric path; a 30° (or safer 40°) mask bounds the loss and QBER.
Across fields: Universal to any ground-to-space link (comms, GNSS, radio astronomy). GNSS often accepts $5\text{--}15^\circ$; QKD wants higher for signal quality.
- **R_{cov} — Ground-coverage radius.** Radius on the ground within which a station can see the satellite above the mask.
This project: 793 km at 30° , 573 km at 40° . **Typical range:** Hundreds of km for LEO.
Why: Sets which city pairs (within $2 \cdot R_{\text{cov}}$) can share one pass — the DUAL/SF split.
Across fields: A coverage-geometry staple; the same footprint idea underlies cellular NTN planning.
- **σ_{R^2} — Rytov variance (scintillation).** Strength of turbulence-induced intensity fluctuations along the path.
This project: < 0.3 across the operating range (weak turbulence). **Typical range:** < 0.3 weak (log-normal fading); $\gtrsim 1$ strong (Gamma-Gamma).
Why: The slant path to LEO stays weak, so log-normal fading applies; the v_3 coefficient is 2.25 (plane-wave slant path).

Across fields: Central to astronomy's 'seeing' and adaptive optics, and to FSO. Fibre and RF do not have optical scintillation.

- **C_n² — Refractive-index structure parameter.** Local turbulence strength versus altitude (Hufnagel-Valley profile).

This project: Ground value $C_n^2(0) = 1.7 \times 10^{-14} \text{ m}^{-2/3}$ (moderate). **Typical range:** Weak 10^{-17} · moderate $\sim 10^{-14}$ · strong $10^{-13} \text{ m}^{-2/3}$.

Why: The ground layer decays within ~ 500 m, so high-altitude wind (not $C_n^2(0)$) dominates the LEO slant-path integral.

Across fields: A term shared by astronomy (site seeing) and FSO; its exotic $\text{m}^{-2/3}$ units come from Kolmogorov turbulence theory.

- **P_T — Transmit optical power.** Average optical power leaving the transmitter.

This project: 1 W (30 dBm). **Typical range:** Satellite laser-comm: ~ 0.1 –5 W.

Why: 1 W is a realistic small-payload figure; higher power eases the link budget but costs SWaP and raises cross-talk (which grows as P^2).

Across fields: Quantum single-photon QKD works at the femtowatt/photon level; this direct-detection SIKD design instead uses a strong classical-style carrier, which is why power is in watts, not photons.

- **m_K — Key-channel modulation index.** Depth of the quantum-key subcarrier's modulation.

This project: 0.15 (corrected from 0.05 in v3). **Typical range:** ~ 0.1 –0.2 feasible at 15 dB filter isolation.

Why: The source SIKD manuscript's feasible region opens for $m_K \gtrsim 0.1$; 0.15 drops QBER below even the 10^{-3} target while leaving budget for data.

Across fields: A subcarrier-multiplexing parameter specific to this SIKD architecture; classical SCM systems tune it purely for SNR, here it also trades against the key's security.

- **m_D — Data-channel modulation index.** Depth of the classical data subcarrier's modulation.

This project: 0.5. **Typical range:** 0.5–0.9 typical; constrained by $m_K + m_D \leq 1$.

Why: Higher m_D raises data throughput but leaks more cross-talk into the key receiver ($\sigma_{CT^2} \propto m_D^2$) — the core SIKD tension.

Across fields: In plain SCM you would push m_D high for capacity; here it is capped because it directly attacks the quantum channel's QBER.

- **Iso — Filter isolation.** How well the RF/optical filter separates the key and data subcarriers.

This project: 15 dB. **Typical range:** ~ 10 –30 dB for practical filters.

Why: Finite isolation lets the data channel leak in as cross-talk; better isolation directly lowers QBER ($\sigma_{CT^2} \propto 10^{-(\text{Iso}/10)}$).

Across fields: A filter-engineering figure; its impact here is amplified because the leaked signal lands on a security-critical quantum channel.

- **R_b — Raw bit rate.** Symbol/bit rate of the link; sets the noise bandwidth $\Delta f = R_b/2$.

This project: 1 Gbps. **Typical range:** Satellite FSO downlinks: Mbps to hundreds of Gbps (TBIRD hit 200 Gbps).

Why: It scales the absolute secret key rate ($\text{SKR}_{\text{bps}} = \text{SKR}_{\text{norm}} \cdot R_b$) and the receiver noise bandwidth.

Across fields: Comms-wide quantity; FSO reaches far higher rates than RF thanks to optical bandwidth.

- **σ_N — Receiver noise (std. dev.).** Total electrical noise: shot + thermal + background + cross-talk.

This project: Sum of four variances; cross-talk dominates when the link is strong. **Typical range:** Set by detector, temperature, bandwidth, and link strength.

Why: It sets the decision thresholds and hence QBER; the v3 cross-talk fix (/8) cut the dominant term four-fold.

Across fields: Every receiver has shot+thermal noise; the cross-talk term is special to this shared-detector SIKD design and, unusually, grows with signal power ($\propto P^2$).

- **σ_{CT²} — Cross-talk noise variance.** Data-channel power leaking into the key receiver: $\propto (R_e P_T m_D h_g h_l)^2 \cdot 10^{-(\text{Iso}/10)/8}$.

This project: Ranges from $\sim 0.5 \times$ to $\sim 290 \times$ thermal noise across elevation. **Typical range:** Grows as received power squared (P^2).

Why: Because it scales as P^2 , it dominates exactly when the link is strongest (high elevation) — a counter-intuitive,

defining SIKD result.

Across fields: Ordinary noise falls or is flat with signal; this term rising with signal is the signature quirk of multiplexing data and key on one detector.

- **QBER — Quantum bit error rate.** Fraction of sifted key bits where Bob disagrees with Alice; the master security indicator.

This project: $< 0.02\%$ across 30° – 90° (v3, $m_K = 0.15$). **Typical range:** Must stay below 11% (BB84) to distil any secret key; stricter protocols want $\leq 10^{-3}$.

Why: Eavesdropping and noise both raise QBER; below threshold a positive secret key survives after error correction and privacy amplification.

Across fields: A QKD-specific metric with a hard security threshold — unlike a classical BER, where 10^{-3} is merely 'poor', here 11% is a wall.

- **SKR — Secret key rate.** Rate of provably-secret key bits: $SKR_{\text{norm}} = P_{\text{sift}}[1 - 2H_2(QBER) - \chi_E]$.

This project: ≈ 61.4 Mbps at 45° elevation (v3). **Typical range:** Zero at $QBER \approx 11\%$; up to tens of Mbps here.

Why: The mutual information minus error-correction leakage minus Eve's information; it collapses steeply near the QBER threshold.

Across fields: The headline output of any QKD system. Real satellite QKD (Micius) delivered kbps; this direct-detection design models Mbps because it uses a strong carrier, not single photons.

- **P_{sift} — Sift probability.** Probability a received sample is confidently decoded (falls outside the dual-threshold guard zone).

This project: Averaged over turbulence by 20-point Gauss-Hermite quadrature. **Typical range:** 0 to 0.5 (half the samples are the key basis).

Why: A wider guard zone lowers errors but sifts out more samples — a yield-vs-error trade set by the threshold scale $\zeta_{\text{scale}} = 2$.

Across fields: Sifting is intrinsic to QKD (basis reconciliation / thresholding); no classical-comms analogue.

- **χ_E — Eve's Holevo information.** The maximum information an eavesdropper can hold about the key.

This project: 0 in the baseline (no eavesdropping assumed). **Typical range:** 0 (no attack) up to values that zero the SKR.

Why: It is what privacy amplification must remove; setting it to 0 characterises performance, not adversarial security.

Across fields: A quantum-information quantity (Holevo bound); has no classical counterpart.

- **H_2 — Binary entropy function.** $H_2(p) = -p \cdot \log_2 p - (1-p) \cdot \log_2 (1-p)$; the information cost of a bit-flip probability p .

This project: $H_2(0.11) \approx 0.4999$. **Typical range:** 0 (at $p=0$ or 1) to 1 (at $p=0.5$).

Why: It appears twice in the SKR bound (mutual information and error-correction leakage), which is why SKR hits zero at $QBER \approx 11\%$.

Across fields: A Shannon-information staple used across coding theory; here it directly sets the QKD security threshold.

- **P_{cloud} / A — Cloud probability & availability.** P_{cloud} is the fraction of hours with thick cloud (link off); availability $A = 1 - P_{\text{cloud}}$.

This project: A ranges 10.7% (Kuala Lumpur, Nov) to 96.5% (Bangkok, Feb). **Typical range:** Wildly location/season dependent in the tropics.

Why: Thick cloud blocks the optical beam entirely, so availability is governed by cloud, not rain; A is invariant to the rain fraction.

Across fields: Optical links treat cloud as total blockage; RF satcom passes through cloud almost unaffected — the fundamental reason tropical optical QKD is hard.

- **c_h — Handover / acquisition cost.** Time lost re-acquiring a satellite when a station switches links.

This project: 30 s charged per handover. **Typical range:** Real FSO acquisition: ~ 10 – 30 s.

Why: Every re-assignment loses delivered key; charging it fairly is why weather-aware scheduling shows a small net cost.

Across fields: FSO needs seconds to re-acquire a narrow beam; RF beam-steering is far faster, so this penalty is an optical-link-specific concern.

9 Index of website implementations (nothing omitted)

The interactive site implements every item below; each is backed by the formulas above.

9.1 Active-recall exercises (90) — 6 sections

#	Section	Items
S1	Channel physics — calculate	18
S2	Geometry & orbits — calculate	12
S3	Weather & availability	12
S4	Key numbers & terms — fill in	15
S5	Committee & literature — match	15
S6	Reasoning, derivations & defense answers	18

Every calc item's numeric answer is computed from the formulas in Sections B–H (never hand-typed); fill/match/order items encode the coefficients, references and reasoning above.

9.2 Multiple-choice questions (360) — 11 professional perspectives

P	Perspective	Items
P1	Quantum Physicist / QKD Theorist	30
P2	FSO Optical-Channel Engineer	30
P3	RF / Receiver / Detection Engineer	30
P4	Orbital-Mechanics / Aerospace Analyst	30
P5	Meteorologist / Climate-Data Scientist	30
P6	Network / Scheduling / Operations Engineer	40
P7	Statistician / Data-Quality Auditor	30
P8	Systems Architect / Integrator	30
P9	Thesis Examiner / Reviewer Angle	30
P10	Applied / Operations / Policy & EDA Breadth	20
P11	Figure, Diagram & Video Reading	60

Perspective 11 (Figure/Video reading) is backed by the result figures; all others by Sections A–H. Full text, answers and one-line theory frameworks are available on the site and in the self-study guides.

Sources

Curated derivations: 05_Code_v3/modules/ and Tu-hoc/SIKD_Satellite_QKD/Study_Materials/ (SIKD_Formula_ Data-driven sections generated from the site's symbols.json, exercises.json, questions.json. Live site: <https://tatcataittn.github.io/for-ttn-USTH/Thesis-M1>.